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Ex.No.4: OFDMA Resource Block Allocation Date

Objective 1:

Code:

clc;

clear;

close all;

% System parameters

bandwidth = 10e6; % 10 MHz

rb_size = 180e3; % 180 kHz per RB

% Total Resource Blocks

total_RB = floor(bandwidth / rb_size);

% Users

users = 1:6;

% RB allocation (equal + remainder distribution)

base = floor(total_RB / length(users));

alloc = base * ones(1,length(users));

```

alloc(1:mod(total_RB,length(users))) = ...
    alloc(1:mod(total_RB,length(users))) + 1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')
disp(total_RB)

disp('RB Allocation per User:')
disp(alloc)

figure

% ----- Graph 1 -----
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

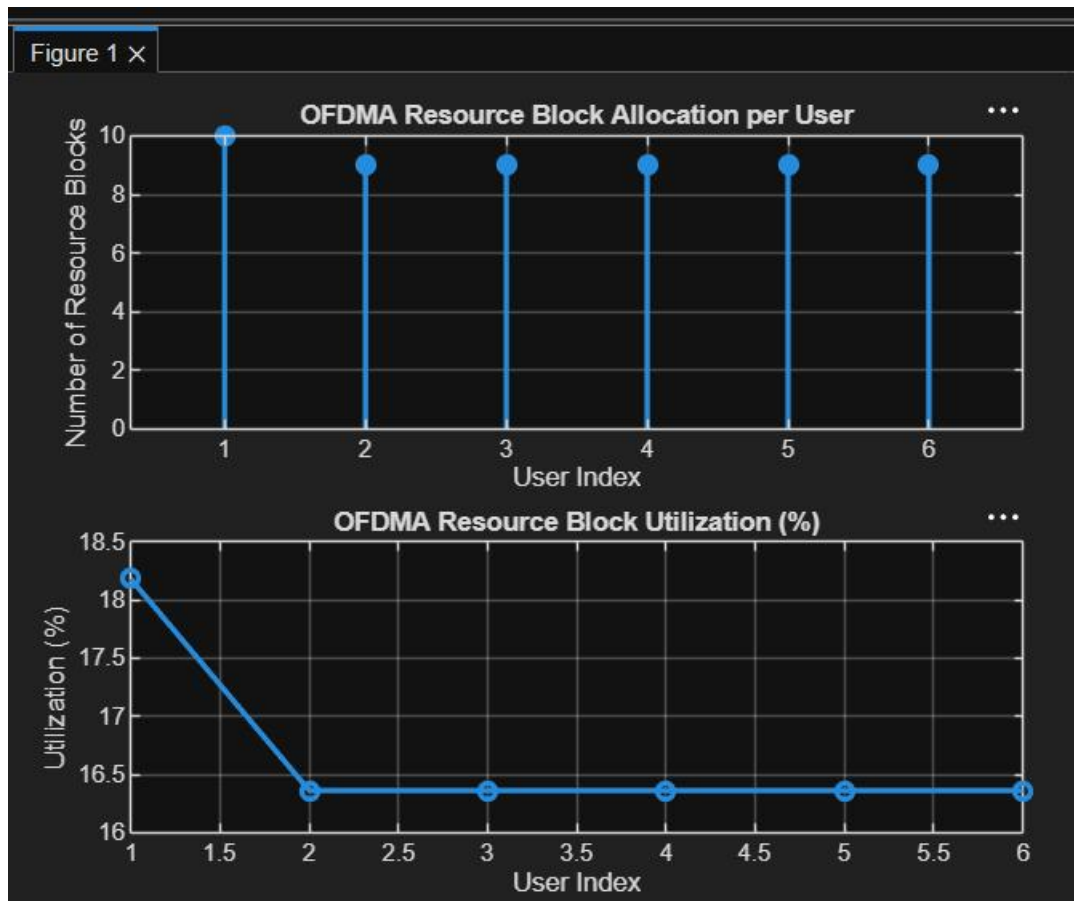
% ----- Graph 2 -----
subplot(2,1,2)

```

```

plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```



OBJECTIVE 2:

CODE:

```

clc; clear; close all;

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

```

```
% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];
% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util)
```

```
figure
```

```
% ----- Graph 1: RB Allocation -----
```

```
subplot(2,1,1)
```

```
plot(users, alloc, '-o', 'LineWidth', 2)
```

```
title('OFDMA Resource Block Allocation per User')
```

```
xlabel('User Index')
```

```
ylabel('Number of Resource Blocks')
```

```
grid on
```

```
% ----- Graph 2: Utilization (%) -----
```

```
subplot(2,1,2)
```

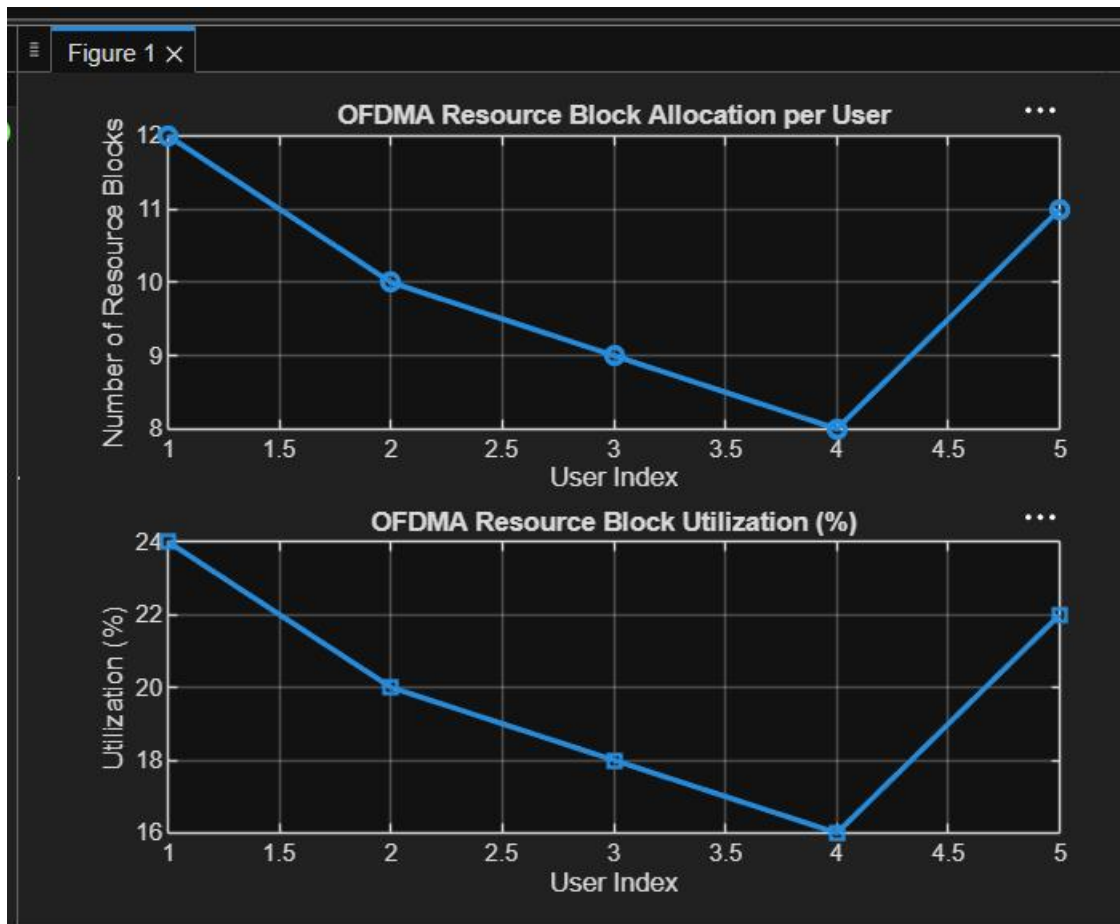
```
plot(users, util, '-s', 'LineWidth', 2)
```

```
title('OFDMA Resource Block Utilization (%)')
```

```
xlabel('User Index')
```

```
ylabel('Utilization (%)')
```

```
grid on
```



OBJECTIVE 3:

CODE:

```
clc; clear; close all;
```

```
% System parameters
```

```
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [12 10 9 8 11];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

```
% Throughput calculation (Mbps)
```

```

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput')

figure

% ----- Graph 1: Throughput per User -----
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on

% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

